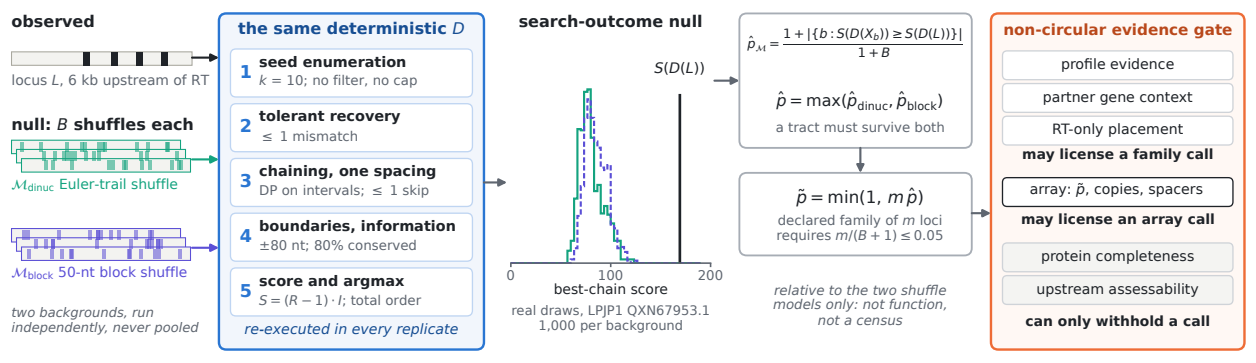
observed
locus L, 6 kb upstream of RT
null: B shuffles each
$\mathcal{M}_{\text{dinuc}}$ Euler-trail shuffle
$\mathcal{M}_{\text{block}}$ 50-nt block shuffle
two backgrounds, run independently, never pooled
the same deterministic D
1 seed enumeration
$k = 10$; no filter, no cap
2 tolerant recovery
$\leq$ 1 mismatch
3 chaining, one spacing
DP on intervals; $\leq$ 1 skip
4 boundaries, information
±80 nt; 80% conserved
5 score and argmax
$S = (R - 1) \cdot I$; total order
re-executed in every replicate
search-outcome null
$S(D(L))$
0
100
200
best-chain score
real draws, LPJP1 QXN67953.1
1,000 per background
$\hat{p}_{\mathcal{M}} = \frac{1 + |\{b : S(D(X_b)) \geq S(D(L))\}|}{1 + B}$
$\hat{p} = \max(\hat{p}_{\text{dinuc}}, \hat{p}_{\text{block}})$
a tract must survive both
$\tilde{p} = \min(1, m\hat{p})$
declared family of m loci
requires $m/(B + 1) \leq 0.05$
relative to the two shuffle models only: not function, not a census
non-circular evidence gate
profile evidence
partner gene context
RT-only placement
may license a family call
array: $\tilde{p}$, copies, spacers
may license an array call
protein completeness
upstream assessability
can only withhold a call